Krishnan R

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.75
Bangalore International Academy	Class XII (Central Board of Secondary Education)	2024	97.60%
Sri Kumarans Childrens Home	Class X (Central Board of Secondary Education)	2022	95.00%

SCHOLASTIC ACHIEVEMENTS

Offered admission to **B.Tech Mathematics & Computing, IISc Bangalore** and **B.Tech CSE, IIIT Bangalore**

NSEJS, NSEP & NSEC: National Top 1%; **NSEA:** State Top 1%

PRMO: Qualified; **NMTC:** Qualified multiple times; **Clash of PI:** Top 13 (India)

Physics Brawl: Rank 1 (Category C)

PROJECTS

RISC-V Toolchain & Web Simulator | C++, WebAssembly, JavaScript, Python

- Engineered a complete RISC-V development toolchain comprising an assembler, disassembler, ISA simulator, and browser-based debugger (**5,691 LOC**, **25+ classes**), enabling end-to-end assembly development and debugging.

- Designed configurable cache simulation supporting multiple replacement and write policies, with an interactive dashboard for comparing cache performance across different hardware configurations.

- Implemented pipelined execution with hazard detection, forwarding, and execution visualization, then compiled the simulator to WebAssembly to provide interactive debugging with breakpoints, step execution, register inspection, and memory visualization.

xv6 Kernel Extensions | C, xv6, RISC-V

- Extended the RISC-V xv6 operating system with **7 custom system calls** for process introspection, parent-child tracking, and system-call monitoring, and replaced the default scheduler with a priority-based multi-level scheduler improving responsiveness while preventing starvation.

- Implemented virtual memory with disk-backed swapping using the Clock page replacement algorithm, alongside per-process memory statistics for debugging and performance analysis.

- Added disk scheduling (FCFS/SSTF) and a RAID 0/1/5 storage simulator across four virtual disks, developing a **5,800+ LOC automated test suite** to validate correctness.

FPU-Assisted Integer Acceleration | C, Gem5, RISC-V, Linux

- Investigated accelerating integer-intensive workloads using idle floating-point hardware, implementing baseline and optimized designs in Gem5 to evaluate architectural trade-offs.

- Achieved **15-20% lower execution time** while validating correctness through architectural simulation.

DistillSpec: Scalable LLM Inference & Experimentation | PyTorch, Hugging Face, Weights & Biases, CUDA

- Collaborated with a Columbia University PhD researcher to develop verifier-aligned speculative decoding techniques that improve LLM inference efficiency through higher draft-model acceptance.

- Built an automated experimentation pipeline managing **60+ training runs** and **3000+ experiments** across A100, H100, and T4 GPUs, enabling reproducible experiment tracking, concurrent execution, and large-scale benchmark evaluation.

- Developed a modular benchmarking framework evaluating **17 speculative decoding methods**, accelerating comparison of training strategies and rapid validation of new research directions.

OCaml Compiler Extension | OCaml, Compiler Design

- Extended an open-source functional language compiler across the lexer, parser, type checker, runtime, and testing framework (**1,064 LOC, 148 commits**), implementing built-in exception handling (raise/try-with) and validating correctness using an automated suite of **68 compiler tests**.

Query Optimization & Performance Analysis | PostgreSQL, SQL, Python

- Benchmarked SQL query performance using **EXPLAIN ANALYZE**, comparing sequential scans, index scans, clustered indexes, and optimizer decisions across varying query selectivities.

- Automated benchmark execution and analyzed PostgreSQL execution plans using Python to study the impact of indexing strategies on query performance.

POSITIONS OF RESPONSIBILITY

Internship Coordinator | Office of Career Services, IIT Hyderabad *(May'26 - Present)*

- Coordinated with recruiters, students, and placement teams to support internship and placement processes, facilitating communication and event execution for campus recruitment activities.

Events Coordinator | NSS, IIT Hyderabad *(May'26 - Present)*

- Planned and executed institute-wide events, coordinating logistics, volunteers, and cross-team communication to ensure smooth end-to-end execution.

RELEVANT COURSES

CS & Mathematics Courses: Introduction to Programming, Software Development Fundamentals, Data Structures, Algorithms, Computer Architecture, Operating Systems, Database Management Systems, Compilers 1

Mathematics Courses: Calculus 1 and 2, Probability and Random Variables, Linear Algebra, Statistics

Other Courses: Digital Fabrication, Digital Circuits, Basic Electrical Engineering

SKILLS

Languages: C, C++, Verilog, RISC-V Assembly, Python, Java, JavaScript, SQL, OCaml

Frameworks/Libraries: PyTorch, NumPy, Hugging Face, Weights & Biases

Tools: Git, Linux, Docker, Gem5, PostgreSQL, Shell, Make